

SEC P vs NP log 2026-05-10

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-10 09:30 UTC

SEC P vs NP — daily log 2026-05-10

24 cycles recorded

Tropical Convex Hull Extreme Points Lower Bound for ACC^0 Circuit Size

- **Verdict:** INCONCLUSIVE
- **Bridge:** Tropical Convex Geometry $\times$ ACC^0 Circuit Size
- **Recorded:** 2026-05-10 00:08 UTC
- **Entry ID:** 0bb43dada997

Statement

For any CNF formula Φ with n variables, the number of extreme points in the tropical convex hull of its clause-indicator polynomial is $\Omega(\log n)$ if Φ has no ACC^0 circuit of size n^c for any constant $c > 0$.

Rationale

Tropical convex hulls encode clause interactions via min-plus geometry, and their extreme points reflect combinatorial constraints on circuit representations. ACC^0 circuits' limited depth imposes structural bounds on these hulls, creating a bridge between algebraic geometry and circuit complexity.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c44b66b7.py", line 78, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c44b66b7.py", line 56, in run_trial
    poly = clause_indicator_polynomial(clauses, n)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c44b66b7.py", line 42, in clause_indica
    poly = [0] * (2 ** n)
           ~~~^~~~~~
```

MemoryError

Judge reasoning

Test crashed with MemoryError before producing results, preventing evaluation of conjecture support/falsification. | next: Optimize clause_indicator_polynomial to handle larger n via sparse representations or incremental construction

Kronecker Coefficient Exponential Gap in Symmetric Powers of Permanent vs Determinant

- **Verdict:** INCONCLUSIVE
- **Bridge:** Representation Theory of Symmetric Groups × Communication Complexity of Disjointness
- **Recorded:** 2026-05-10 00:22 UTC
- **Entry ID:** 722bbadc85c5

Statement

For all $n \leq 40$, the Kronecker coefficient for the decomposition of $\text{Sym}^k(\text{Sym}_n(V))$ into irreducible representations is exponentially larger for the permanent than for the determinant when $k = \lceil n/2 \rceil$, implying a $\Omega(n)$ communication complexity lower bound for the corresponding disjointness instance.

Rationale

Kronecker coefficients capture the structure of symmetric power decompositions, which are central to GCT's orbit closure separations. Linking them to communication complexity provides a bridge between algebraic geometry and combinatorial lower bounds, leveraging the exponential sensitivity of symmetric tensors to combinatorial constraints.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
nterexample': 'Ratio 1.0 is not exponentially larger than 2^7'}
TRIAL: {'metric_name': 'Kronecker Coefficient Ratio', 'metric_value': 1.0, 'instances_tested': 1, 'co
TRIAL: {'metric_name': 'Kronecker Coefficient Ratio', 'metric_value': 1.0, 'instances_tested': 1, 'co
TRIAL: {'metric_name': 'Kronecker Coefficient Ratio', 'metric_value': 1.0, 'instances_tested': 1, 'co
TRIAL: {'metric_name': 'Kronecker Coefficient Ratio', 'metric_value': 1.0, 'instances_tested': 1, 'co
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TRIAL: {'metric_name': 'Kronecker Coefficient Ratio', 'metric_value': 1.0, 'instances_tested': 1, 'co
TRIAL: {'metric_name': 'Kronecker Coefficient Ratio', 'metric_value': 1.0, 'instances_tested': 1, 'co
TRIAL: {'metric_name': 'Kronecker Coefficient Ratio', 'metric_value': 1.0, 'instances_tested': 1, 'co
TRIAL: {'metric_name': 'Kronecker Coefficient Ratio', 'metric_value': 1.0, 'instances_tested': 1, 'co
TRIAL: {'metric_name': 'Kronecker Coefficient Ratio', 'metric_value': 1.0, 'instances_tested': 1, 'co
```

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4e40f8cb.py", line 81, in <module>
    first_failing_seed = next(r["seed"] for r in res
```

Judge reasoning

Test crashed before completing evaluation, but partial results show counterexamples where ratio 1.0 fails to meet exponential thresholds. | next: Fix seed handling in test code and re-run with full coverage of $n \leq 40$ cases

Operator Norm Lower Bound for Disjointness Communication Matrices

- **Verdict:** INCONCLUSIVE
- **Bridge:** Operator Spaces Theory $\times$ Communication Complexity of DISJOINTNESS
- **Recorded:** 2026-05-10 00:30 UTC
- **Entry ID:** 4da9869ae44e

Statement

For any $n \times n$ matrix M representing the DISJOINTNESS function, the completely bounded norm $\|M\|_{cb} \geq \Omega(n)$. This holds for all matrices where $M[i,j] = 1$ iff the i -th and j -th subsets are disjoint, with random subset generation using uniform distribution over $2^{\lceil \log_2 n \rceil}$ -element sets.

Rationale

Operator spaces theory provides tools to analyze tensor norms that capture the structure of communication matrices. The cb -norm, being invariant under tensor products, could expose inherent dimensionality constraints in DISJOINTNESS that classical discrepancy methods miss.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dc72256d.py", line 78, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dc72256d.py", line 60, in run_trial
    cb_norm_max, _ = numerical_range(M)
                     ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dc72256d.py", line 46, in numerical_range
    B = [M[i] - (i+1)*I for i in range(n)]
         ~~~~^~~~~~
```

TypeError: unsupported operand type(s) for -: 'list' and 'list'

Judge reasoning

Test crashed due to TypeError in matrix operations, preventing result collection | next: Fix the numerical_range function's matrix subtraction logic to handle list operations properly

Tensor Rank Exponential Gap in Read-Twice vs Read-Once Branching Programs for IP₂

- **Verdict:** INCONCLUSIVE
- **Bridge:** Tensor Rank over Real Numbers $\times$ Read-Twice Branching Programs
- **Recorded:** 2026-05-10 00:53 UTC
- **Entry ID:** ec8a0368610a

Statement

For any read-once branching program P over n variables, the tensor rank of the transition tensor $T(P)$ is $O(\log n)$. For the IP_2 function, any read-twice branching program Q must satisfy $\text{rank}(T(Q)) \geq 2^{\Omega(n/2)} / \text{poly}(n)$.

Rationale

Tensor rank captures the algebraic complexity of branching program transitions. Read-once BPs decompose as low-rank tensors, while read-twice BPs require high-rank tensors to model their repeated access to variables, mirroring the exponential gap in communication complexity for IP 2.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

- -STDERR- -

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6ec1712e.py", line 135, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6ec1712e.py", line 90, in run_trial
    read_once_matrix = matrix_multiplication(*read_once_bp)
                                ~~~~~^~~~~~
```

```
TypeError: matrix_multiplication() takes 2 positional arguments but 5 were given
```

Judge reasoning

Test crashed due to incorrect argument count in matrix multiplication, preventing result collection
| next: Fix matrix multiplication argument count in test_6ec1712e.py

Projective Plane Line Count Bounds ABP Width for Monotone Circuits

- **Verdict:** INCONCLUSIVE
- **Bridge:** Finite Geometry $\times$ Monotone Boolean Circuit Size
- **Recorded:** 2026-05-10 01:02 UTC
- **Entry ID:** 96cc863111fe

Statement

For every monotone Boolean circuit computing the characteristic function of a projective plane $\text{PG}(2, q)$, the minimal width of an ABP computing the same function satisfies $\text{width} \geq q^2 + q + 1$. This bound is tight for $q = 2^k$.

Rationale

Projective planes exhibit highly symmetric incidence structures that could encode combinatorial constraints mirroring the layered computation of ABPs. The tightness of the bound suggests a deep algebraic relationship between geometric incidence and algebraic computation.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_55481e38.py", line 112, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_55481e38.py", line 80, in run_trial
    points, lines = generate_projective_plane(q)
                    ^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_55481e38.py", line 20, in generate_proj
    raise ValueError("q must be at least 2")
ValueError: q must be at least 2
```

Judge reasoning

Test crashed due to invalid q value (must be ≥ 2), preventing evaluation of conjecture | next: Run test with q=2 and valid projective plane parameters to verify bounds

Quantum Rank Lower Bounds for Disjointness Communication Matrices

- **Verdict:** INCONCLUSIVE
- **Bridge:** Noncommutative Geometry $\times$ Communication Complexity of DISJOINTNESS
- **Recorded:** 2026-05-10 01:08 UTC
- **Entry ID:** 8dbbdbbc746a

Statement

For the $n \times n$ DISJOINTNESS matrix M_n , the noncommutative rank $\tau(M_n)$ satisfies $\tau(M_n) \geq c \cdot n$ for some universal constant $c > 0$. This invariant is defined as the minimal k such that M_n can be expressed as a sum of k rank-1 matrices in the matrix algebra over the noncommutative ring $\mathbb{C}\langle x, y \rangle / (x^2 = x, y^2 = y)$.

Rationale

Noncommutative geometry provides tools to analyze matrix factorizations through ring-theoretic structures. The DISJOINTNESS matrix's inherent combinatorial structure may enforce a nontrivial lower bound on its noncommutative rank, revealing algebraic obstructions to efficient communication protocols.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_82a6913d.py", line 91, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_82a6913d.py", line 67, in run_trial
    tau_M_n = noncommutative_rank(M_n, n)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_82a6913d.py", line 58, in noncommutative_rank
    A = [[x[i][j] - y[j] * M[i][n + j] for j in range(n)] for i in range(n)]
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

TypeError: unsupported operand type(s) for -: 'int' and 'list'

Judge reasoning

Test crashed due to TypeError, preventing result collection. Pre-registered support condition cannot be evaluated without successful trials. | next: Debug the noncommutative_rank function's matrix construction to resolve type mismatches in the algebraic operations

Symmetric Tensor Rank Gap in Permanent vs Determinant Decompositions

- **Verdict:** INCONCLUSIVE
- **Bridge:** Representation Theory of Symmetric Groups $\times$ Monotone Circuit Size for 3-CNFs
- **Recorded:** 2026-05-10 01:23 UTC
- **Entry ID:** 8de14a1366b2

Statement

For a random 3-CNF formula Φ with n variables, the dimension of the irreducible constituent λ in the symmetric square of the permanent's representation (computed via Young diagrams) exceeds the dimension of the corresponding constituent in the determinant's representation by a factor of $\Omega(2^{\{n/4\}})$ with probability ≥ 0.95 over Φ .

Rationale

Symmetric tensor decompositions reveal structural asymmetry between permanent and determinant. The Young diagram dimensions capture combinatorial complexity, and their exponential gap could imply inherent circuit size separation via representation-theoretic obstructions.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6dfe40e5.py", line 110, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6dfe40e5.py", line 83, in run_trial
    perm_val = permanent(n, perm_matrix)
    ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6dfe40e5.py", line 42, in permanent
    submatrix = [[matrix[i][k] for k in range(j) + range(j+1, n)] for i in range(1, n)]
                  ~~~~~^~~~~~
```

TypeError: unsupported operand type(s) for +: 'range' and 'range'

Judge reasoning

Test crashed due to Python TypeError in range concatenation, preventing data collection. Support condition (80% success rate) cannot be evaluated without valid results. | next: Fix the range concatenation error in the permanent function by using list concatenation (e.g., list(range(j)) + list(range(j+1, n)))

Negative Eigenvalue Count Lower Bounds SOS Degree for Max-CUT

- **Verdict:** INCONCLUSIVE
- **Bridge:** Real Algebraic Geometry × SOS Degree for Max-CUT
- **Recorded:** 2026-05-10 01:47 UTC
- **Entry ID:** 510fd15279df

Statement

For any n -variable Max-CUT instance, the minimal SOS degree d required to certify its integrality gap satisfies $d \geq k + 1$, where k is the number of negative eigenvalues of the moment matrix of the degree- d pseudoexpectation.

Rationale

The number of negative eigenvalues quantifies the ‘deficiency’ in positive semidefiniteness of the moment matrix, which directly correlates with the need for higher-degree SOS constraints to resolve non-positivity in the polynomial representation of the Max-CUT objective.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.14s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0ee3b164.py", line 116, in <module>
    result = run_trial(seed)
    ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0ee3b164.py", line 91, in run_trial
    eigenvalues = [abs(e) for e in gaussian_elimination(M)[-1]]
                  ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0ee3b164.py", line 28, in gaussian_elim
    factor = 1 / A[i][i]
    ~~~~~
```

ZeroDivisionError: division by zero

Judge reasoning

Test crashed with ZeroDivisionError, preventing data collection. Pre-registered support condition cannot be evaluated without valid results. | next: Debug the gaussian_elimination function to handle singular matrices, then re-run tests with multiple seeds

Plethysm Coefficient Gap in Homogeneous Polynomial Decompositions

- **Verdict:** INCONCLUSIVE
- **Bridge:** Plethysm Theory $\times$ Boolean Circuit Size
- **Recorded:** 2026-05-10 03:39 UTC
- **Entry ID:** cc61e4e0da28

Statement

For any $n \geq 2$, the plethysm coefficient of the permanent polynomial perm_n decomposed into symmetric squares of $\text{GL}(n)$ representations exceeds that of det_m for all $m < n^{\{1.5\}}$, with ratio $\geq 2^{\{\Omega(n)\}}$

Rationale

Plethysm coefficients encode symmetry-breaking patterns in polynomial representations. If permanent's decomposition requires exponentially more symmetric squares than determinant's, this could expose structural asymmetries exploitable for circuit lower bounds.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 124, elapsed: 240.00s

TIMEOUT after 240s

Judge reasoning

Test timed out before producing results, preventing verification of support fraction or counterexamples. | next: Run test with extended timeout and higher resource allocation

Hypergraph Maximum Matching Inverse Proportional to ACC^0 Circuit Size for 3-CNFs

- **Verdict:** INCONCLUSIVE
- **Bridge:** Hypergraph Theory $\times$ ACC^0 Circuit Size
- **Recorded:** 2026-05-10 03:58 UTC
- **Entry ID:** 9d4ad01564fe

Statement

For any 3-CNF formula with n variables, the maximum matching size of its clause hypergraph is $\Theta(\log n)$ if and only if the formula can be computed by an ACC^0 circuit of size $\Theta(n^k)$ for some constant k .

Rationale

The maximum matching size captures the sparsity of clause overlaps. Sparse overlaps (logarithmic) suggest a structured dependency that can be captured by ACC^0 circuits, which have limited depth but can handle structured patterns.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.10s

```
'max_matching_size', 'metric_value': 85, 'instances_tested': 1, 'conjecture_holds': False, 'countere
TRIAL: {'metric_name': 'max_matching_size', 'metric_value': 23, 'instances_tested': 1, 'conjecture_h
TRIAL: {'metric_name': 'max_matching_size', 'metric_value': 153, 'instances_tested': 1, 'conjecture_
TRIAL: {'metric_name': 'max_matching_size', 'metric_value': 143, 'instances_tested': 1, 'conjecture_
TRIAL: {'metric_name': 'max_matching_size', 'metric_value': 38, 'instances_tested': 1, 'conjecture_h
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TRIAL: {'metric_name': 'max_matching_size', 'metric_value': 133, 'instances_tested': 1, 'conjecture_
TRIAL: {'metric_name': 'max_matching_size', 'metric_value': 124, 'instances_tested': 1, 'conjecture_
```

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_70a9fc8c.py", line 82, in <module>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                        ~~~~~~
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_70a9fc8c.py", line 82, in <genexpr>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                        ~~~~~~
```

KeyError: 'seed'

Judge reasoning

Test crashed before completing validation; counterexamples in output may be unreliable | next:
Re-run test with increased time limits and deterministic seed configuration

Finite Field Rank and Branching Program Width for Boolean Functions

- **Verdict:** INCONCLUSIVE
- **Bridge:** Finite Field Linear Algebra $\times$ Branching Program Width
- **Recorded:** 2026-05-10 04:08 UTC
- **Entry ID:** 89d7e3ddf0de

Statement

For a Boolean function f on n variables, let M_f be the matrix over $GF(2)$ with rows representing variables and columns representing truth assignments, where $M_f[i][j] = f(x_j)$ if x_j has i -th variable set. The minimal branching program width for f is at least the rank of M_f over $GF(2)$.

Rationale

The rank of M_f captures the linear dependencies in f 's truth table. A higher rank implies more complex interactions between variables, necessitating wider branching programs to represent the function's behavior. This link could expose algebraic structure in complexity measures.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.10s

```
TRIAL: {'metric_name': 'rank vs. BDD width', 'metric_value': 10, 'instances_tested': 1, 'conjecture_h  
TRIAL: {'metric_name': 'rank vs. BDD width', 'metric_value': 10, 'instances_tested': 1, 'conjecture_h
```

```
--STDERR--
```

```
Traceback (most recent call last):
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b46512cb.py", line 72, in <module>  
    result = run_trial(seed)  
            ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b46512cb.py", line 56, in run_trial  
    bdd_width_value = bdd_width(M_f)  
                      ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b46512cb.py", line 45, in bdd_width  
    if truth_table[i][var] != truth_table[2 * i][var]:  
                                   ~~~~~^~~~~~
```

```
IndexError: list index out of range
```

Judge reasoning

Test crashed with IndexError during BDD width calculation, preventing reliable results | next: Fix the index out-of-range error in bdd_width function and re-run experiments

Matroid Rank Inverse Proportional to ACC^0 Circuit Size for GF(2) Polynomials

- **Verdict:** INCONCLUSIVE
- **Bridge:** Matroid Theory $\times$ ACC^0 Circuit Size
- **Recorded:** 2026-05-10 04:25 UTC
- **Entry ID:** 74050dd8b5ed

Statement

For any GF(2)-polynomial f over n variables, the rank r of its matroid representation satisfies $size_{\{ACC^0\}}(f) \geq \Omega(n^{\{1 + 1/r\}})$

Rationale

Matroid rank captures combinatorial dependencies in GF(2) polynomials, while ACC^0 circuit size reflects algebraic complexity. Their inverse proportionality could reveal structural constraints on circuit depth for dependency-rich functions.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 124, elapsed: 240.37s

TIMEOUT after 240s

Judge reasoning

Test timed out before producing results; insufficient data to evaluate support fraction or counterexamples | next: Run test with extended timeout and higher resource allocation

Secant Variety Dimension Lower Bounds Disjointness Communication Complexity

- **Verdict:** INCONCLUSIVE
- **Bridge:** Secant Variety Geometry $\times$ Communication Complexity of DISJOINTNESS
- **Recorded:** 2026-05-10 04:55 UTC
- **Entry ID:** 059c72c72c3c

Statement

For a random $n \times n$ DISJOINTNESS matrix M , the dimension of its secant variety satisfies $\dim(\sigma(M)) \geq \Omega(n)$. This holds for all instances where the communication matrix has no zero rows or columns.

Rationale

Secant variety dimensions capture the algebraic structure of communication matrices. For DISJOINTNESS, the high-dimensional geometry of its matrix (which is a 0-1 matrix with row/column support constraints) may enforce a lower bound on the secant variety's dimension, mirroring the $\Omega(n)$ communication complexity lower bound.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_928e789b.py", line 75, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_928e789b.py", line 54, in run_trial
    dim_secant_variety = matrix_rank(M)
                        ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_928e789b.py", line 41, in matrix_rank
    return gaussian_elimination(A)
           ^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_928e789b.py", line 31, in gaussian_elimination
    A[j][i] /= factor
```

ZeroDivisionError: division by zero

Judge reasoning

Test crashed with ZeroDivisionError, preventing evaluation of the conjecture's validity. | next: Fix numerical stability in Gaussian elimination to handle singular matrices

Real Critical Point Count Exponential in SOS Degree for Max-CUT

- **Verdict:** INCONCLUSIVE
- **Bridge:** Real Algebraic Geometry $\times$ SOS Degree for Max-CUT
- **Recorded:** 2026-05-10 05:06 UTC
- **Entry ID:** 63ccbc681743

Statement

For a random Max-CUT instance with n variables, the number of real critical points of the quartic polynomial derived from the instance is exponential in the SOS degree d required to achieve an approximation ratio of $0.878 - \epsilon$.

Rationale

The number of real critical points reflects the complexity of the optimization landscape. The SOS hierarchy must capture these points to approximate the true minimum, implying a higher degree for more complex landscapes.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1a142126.py", line 123, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1a142126.py", line 98, in run_trial
    critical_points_count = len(critical_points(quartic_poly))
                             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1a142126.py", line 84, in critical_points
    n = len(quartic_polynomial)
        ^^^^^^^^^^^^^^^^^^^^^
```

TypeError: object of type 'Fraction' has no len()

Judge reasoning

Test crashed due to TypeError in critical points calculation; no data produced to evaluate conjecture | next: Fix the Fraction object handling in critical_points() function and rerun experiments

Schatten p-Norm Lower Bound for Disjointness Communication Complexity

- **Verdict:** INCONCLUSIVE
- **Bridge:** Noncommutative L^p Geometry $\times$ Communication Complexity of DISJOINTNESS
- **Recorded:** 2026-05-10 05:57 UTC
- **Entry ID:** 835e60ff99be

Statement

For any $n \times n$ matrix M representing a DISJOINTNESS instance, the Schatten p -norm $\|M\|_p \geq \Omega(\sqrt{n})$ for $p=2$. This bound is tight for the standard DISJOINTNESS matrix and holds for all matrices with $\text{rank} \leq n$.

Rationale

Noncommutative L^p norms capture operator-theoretic structure of tensor products, which may encode the entanglement required for communication. The Schatten $p=2$ norm (Frobenius) relates to the Hilbert-Schmidt energy of the matrix, potentially mirroring the information-theoretic cost of coordination.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.03s

```
unterexample': ''
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 6.324555320336759, 'instances_tested': 1, '
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 6.324555320336759, 'instances_tested': 1, '
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 6.324555320336759, 'instances_tested': 1, '
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 6.324555320336759, 'instances_tested': 1, '
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 6.324555320336759, 'instances_tested': 1, '
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 6.324555320336759, 'instances_tested': 1, '
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 6.324555320336759, 'instances_tested': 1, '
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 6.324555320336759, 'instances_tested': 1, '
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TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 6.324555320336759, 'instances_tested': 1, '
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 6.324555320336759, 'instances_tested': 1, '
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 6.324555320336759, 'instances_tested': 1, '
RESULT: SUPPORTED mean=6.324555320336759 std=0.0 support_fraction=1.0
```

Judge reasoning

Single instance tested ($n=40$) meets the claimed lower bound exactly, but insufficient to establish asymptotic scaling. The metric value $6.3245 \approx \sqrt{40}$ suggests a constructed test case rather than empirical validation of $\Omega(\sqrt{n})$ for growing n . | next: Test with larger n (e.g., $n=1000$) to observe scaling behavior and verify the lower bound holds asymptotically

SOS Degree Lower Bound via Adjacency Eigenvalue Multiplicity

- **Verdict:** INCONCLUSIVE
- **Bridge:** Real Algebraic Geometry $\times$ SOS Refutation Degree for Max-CUT

- **Recorded:** 2026-05-10 06:28 UTC
- **Entry ID:** 8131a890bdd5

Statement

For any n -vertex graph G , the SOS refutation degree $d(G)$ required to certify $\text{Max-CUT}(G) \leq \alpha$ satisfies $d(G) \geq k(G)$, where $k(G)$ is the number of distinct positive eigenvalues of G 's adjacency matrix. Equality holds for complete bipartite graphs.

Rationale

The Positivstellensatz provides certificates for non-negativity of polynomials, which underlie SOS proofs. The multiplicity of positive eigenvalues reflects the algebraic structure of the quadratic form defining Max-CUT, creating a bridge between spectral geometry and proof complexity.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ldce4a38.py", line 108, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ldce4a38.py", line 89, in run_trial
    eigenvals = eigenvalues(A)
                ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ldce4a38.py", line 55, in eigenvalues
    Q, R = qr_decomposition(A)
            ^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ldce4a38.py", line 70, in qr_decomposition
    Q[k] = [v[i] - norm_v * e_k[i] for i in range(n)]
           ~^^^
```

IndexError: list index out of range

Judge reasoning

Test crashed during eigenvalue computation, preventing validation of conjecture | next: Fix the QR decomposition implementation to handle edge cases in eigenvalue calculation

Additive Energy Lower Bound via Read-Twice Branching Program Discrepancy

- **Verdict:** INCONCLUSIVE
- **Bridge:** Additive Combinatorics × Read-Twice Branching Programs
- **Recorded:** 2026-05-10 07:07 UTC
- **Entry ID:** 1de2d7187fe1

Statement

For every read-twice branching program P computing a function $f: \{0,1\}^n \rightarrow \{-1,1\}$, the additive energy $E(f)$ of f is at least $\Omega(n)$ times the discrepancy $\text{disc}(P)$ of P .

Rationale

Additive energy measures the number of additive configurations in a function, which could expose structural constraints on functions computable by read-twice BPs. If read-twice BPs have limited additive structure, their discrepancy (a measure of imbalance) might inversely correlate with additive energy, suggesting a nontrivial relationship between algebraic structure and computational complexity.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 124, elapsed: 240.07s

TIMEOUT after 240s

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Run the test with extended time limits and verify hardware/software constraints

Real Root Count Exponential in AC^0 PARITY Circuits

- **Verdict:** INCONCLUSIVE
- **Bridge:** Real Algebraic Geometry $\times$ AC^0 Circuit Size for PARITY
- **Recorded:** 2026-05-10 07:15 UTC
- **Entry ID:** a1e4626e2b6a

Statement

For any AC^0 circuit C computing PARITY on n variables, the number of real roots of the polynomial P_C derived from C is $\Theta(2^{\lfloor n/2 \rfloor})$

Rationale

The parity function's Fourier transform has specific properties, and the number of real roots of P_C could reflect the circuit's structure, potentially exposing lower bounds via real algebraic geometry

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5fd1301b.py", line 88, in <module>
    result = run_trial(seed)
              ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5fd1301b.py", line 62, in run_trial
    poly = cnf_to_poly(cnf)
            ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5fd1301b.py", line 48, in cnf_to_poly
    term *= (1 - x**literal)
              ^
NameError: name 'x' is not defined
```

Judge reasoning

Test crashed due to undefined variable 'x' in polynomial construction, preventing evaluation of root counts. | next: Fix the polynomial construction code to define variables properly and re-run tests

Polymatroid Rank Lower Bound for Monotone CLIQUE

- **Verdict:** INCONCLUSIVE
- **Bridge:** Polymatroid Theory × Monotone Circuit Size for k-CLIQUE
- **Recorded:** 2026-05-10 07:36 UTC
- **Entry ID:** 834e913154a8

Statement

For any graph G on n vertices, let $M(G)$ be the polymatroid defined by the rank function $r(S) = \max\{|S \cap C| : C \text{ is a clique in } G\}$. Then $r(M(G)) \geq \Omega(n^{\{1/2\}})$ for all graphs G with clique number $\leq n^{\{1/2\}} + 1$.

Rationale

Polymatroid rank captures submodular constraints inherent in clique enumeration. By linking this geometric invariant to monotone circuit complexity, we expose structural limitations in representing clique problems with AND/OR gates, potentially revealing new barriers for monotone lower bounds.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 124, elapsed: 240.10s

TIMEOUT after 240s

Judge reasoning

Test timed out without producing results, preventing evaluation of support fraction or counterexamples. | next: Run test with extended timeout and higher n values to gather conclusive data

Modular Character Degree Gap in Permanent vs Determinant Circuits

- **Verdict:** INCONCLUSIVE
- **Bridge:** Modular Representation Theory × Monotone Circuit Size for Permanent
- **Recorded:** 2026-05-10 07:48 UTC
- **Entry ID:** dd4699216f41

Statement

For a random 3-CNF formula Φ with n variables, let d be its minimal degree of an irreducible representation of the symmetric group S_n over $\text{GF}(2)$. Then, the monotone circuit size of the permanent polynomial over $\text{GF}(2)$ is at least $\Omega(2^{\{d\}} / n^2)$.

Rationale

Modular representation theory of symmetric groups provides invariants that could distinguish algebraic structures of permanent and determinant. The degree of irreducible representations relates to symmetry-breaking patterns, which might encode computational complexity via combinatorial constraints on monotone circuits.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.47s

```
re_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'circuit_size_over_bound', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'circuit_size_over_bound', 'metric_value': 28.125, 'instances_tested': 1, 'conjecture_holds': False}
TRIAL: {'metric_name': 'circuit_size_over_bound', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'circuit_size_over_bound', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'circuit_size_over_bound', 'metric_value': 9.375, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'circuit_size_over_bound', 'metric_value': 9.375, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'circuit_size_over_bound', 'metric_value': 12.5, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'circuit_size_over_bound', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'circuit_size_over_bound', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': True}
```

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_51da1542.py", line 134, in <module>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                           ~~~~~^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_51da1542.py", line 134, in <genexpr>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                           ~~~~~^
```

KeyError: 'seed'

Judge reasoning

Test crashed before completing all trials; incomplete results prevent evaluation of support fraction or counterexamples. | next: Re-run test with debug logging to capture full results and ensure 'seed' metadata is properly recorded.

Class Number Inverse Proportional to Resolution Length in 3-SAT

- **Verdict:** INCONCLUSIVE
- **Bridge:** Algebraic Number Theory $\times$ Resolution Proof Length
- **Recorded:** 2026-05-10 08:11 UTC
- **Entry ID:** 3758fc6a2d53

Statement

For a 3-SAT instance with n variables, the class number $h(K)$ of the number field $K = \mathbb{Q}(\sqrt[m]{m})$ (where m is the number of clauses) satisfies $h(K) \geq c * (\log n) / L$, where L is the minimal resolution proof length. Equality holds when the instance is a random 3-SAT formula with density 2.5.

Rationale

Algebraic number theory's class number captures arithmetic complexity of number fields, which may mirror structural properties of SAT instances. The inverse proportionality suggests that harder-to-prove formulas correspond to fields with smaller class numbers, revealing a hidden arithmetical connection.

Novelty

- Judge: INCONCLUSIVE over 10 arXiv hits

Top hits consulted: - [1908.05361v2] Placing quantified variants of 3-SAT and Not-All-Equal 3-SAT in the polynomial hierarchy - [2202.08382v3] The relative class number one problem for function fields, I - [1307.5776v2] An exact algorithm for 1-in-3 SAT - [2402.07935v3] On the Frobenius fields of abelian varieties over number fields - [1803.08437v5] The étale cohomology ring of the ring of integers of a number field

Empirical Test

- exit code: 1, elapsed: 3.56s

```
: 0.2710850306818168, 'instances_tested': 30, 'conjecture_holds': False, 'counterexample': 'Counterexample metric suggests falsification, but incomplete results invalidate definitive conclusion. | next: Re-run test with seed tracking enabled and extended instance count to validate conjecture across complete dataset'
TRIAL: {'metric_name': 'h(K) * L / log(n)', 'metric_value': 0.2710850306818168, 'instances_tested': 30, 'conjecture_holds': False, 'counterexample': 'Counterexample metric suggests falsification, but incomplete results invalidate definitive conclusion. | next: Re-run test with seed tracking enabled and extended instance count to validate conjecture across complete dataset'}
TRIAL: {'metric_name': 'h(K) * L / log(n)', 'metric_value': 0.2710850306818168, 'instances_tested': 30, 'conjecture_holds': False, 'counterexample': 'Counterexample metric suggests falsification, but incomplete results invalidate definitive conclusion. | next: Re-run test with seed tracking enabled and extended instance count to validate conjecture across complete dataset'}
TRIAL: {'metric_name': 'h(K) * L / log(n)', 'metric_value': 0.2710850306818168, 'instances_tested': 30, 'conjecture_holds': False, 'counterexample': 'Counterexample metric suggests falsification, but incomplete results invalidate definitive conclusion. | next: Re-run test with seed tracking enabled and extended instance count to validate conjecture across complete dataset'}
TRIAL: {'metric_name': 'h(K) * L / log(n)', 'metric_value': 0.2710850306818168, 'instances_tested': 30, 'conjecture_holds': False, 'counterexample': 'Counterexample metric suggests falsification, but incomplete results invalidate definitive conclusion. | next: Re-run test with seed tracking enabled and extended instance count to validate conjecture across complete dataset'}
TRIAL: {'metric_name': 'h(K) * L / log(n)', 'metric_value': 0.2710850306818168, 'instances_tested': 30, 'conjecture_holds': False, 'counterexample': 'Counterexample metric suggests falsification, but incomplete results invalidate definitive conclusion. | next: Re-run test with seed tracking enabled and extended instance count to validate conjecture across complete dataset'}
TRIAL: {'metric_name': 'h(K) * L / log(n)', 'metric_value': 0.2710850306818168, 'instances_tested': 30, 'conjecture_holds': False, 'counterexample': 'Counterexample metric suggests falsification, but incomplete results invalidate definitive conclusion. | next: Re-run test with seed tracking enabled and extended instance count to validate conjecture across complete dataset'}
TRIAL: {'metric_name': 'h(K) * L / log(n)', 'metric_value': 0.2710850306818168, 'instances_tested': 30, 'conjecture_holds': False, 'counterexample': 'Counterexample metric suggests falsification, but incomplete results invalidate definitive conclusion. | next: Re-run test with seed tracking enabled and extended instance count to validate conjecture across complete dataset'}
TRIAL: {'metric_name': 'h(K) * L / log(n)', 'metric_value': 0.2710850306818168, 'instances_tested': 30, 'conjecture_holds': False, 'counterexample': 'Counterexample metric suggests falsification, but incomplete results invalidate definitive conclusion. | next: Re-run test with seed tracking enabled and extended instance count to validate conjecture across complete dataset'}
TRIAL: {'metric_name': 'h(K) * L / log(n)', 'metric_value': 0.2710850306818168, 'instances_tested': 30, 'conjecture_holds': False, 'counterexample': 'Counterexample metric suggests falsification, but incomplete results invalidate definitive conclusion. | next: Re-run test with seed tracking enabled and extended instance count to validate conjecture across complete dataset'}
```

Judge reasoning

Test crashed before completing data collection, preventing evaluation of support fraction. Counterexample metric suggests falsification, but incomplete results invalidate definitive conclusion. | next: Re-run test with seed tracking enabled and extended instance count to validate conjecture across complete dataset

BIBD Incidence Matrix ACC⁰ Circuit Size Lower Bound

- **Verdict:** INCONCLUSIVE
- **Bridge:** Combinatorial Designs (Balanced Incomplete Block Designs) $\times$ ACC⁰ Circuit Size

- **Recorded:** 2026-05-10 08:23 UTC
- **Entry ID:** f7b05c9ca4f6

Statement

For any Balanced Incomplete Block Design (BIBD) with parameters (v, k, λ) , the corresponding incidence matrix function $f: \{0,1\}^v \rightarrow \{0,1\}$ requires ACC⁰ circuits of size $\Omega(v^{2 - 1/k})$.

Rationale

The regular structure of BIBDs imposes constraints on the incidence matrix that make it resistant to efficient ACC⁰ computation, leading to a lower bound on circuit size.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 124, elapsed: 240.06s

TIMEOUT after 240s

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Increase timeout duration and re-run test with extended computation limits

Minimal Incidence Count Inverse Proportional to Disjointness Communication Complexity

- **Verdict:** INCONCLUSIVE
- **Bridge:** Incidence Geometry $\times$ Communication Complexity of Disjointness
- **Recorded:** 2026-05-10 09:07 UTC
- **Entry ID:** bee7b259537e

Statement

For a disjointness function f on n variables, the minimal number of incidences between the points and lines in the corresponding incidence structure is inversely proportional to the communication complexity of f , i.e., $\log(\text{communication complexity}) = \Theta(1 / \min_incidence)$.

Rationale

Incidence geometry provides tools to analyze geometric configurations, which might reveal structural properties of communication matrices that influence their complexity. The inverse proportionality suggests that more structured configurations (higher $\min_incidence$) could imply lower communication complexity.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f4380202.py", line 88, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f4380202.py", line 52, in run_trial
    comm_complexity = compute_communication_complexity(points, lines)
                    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f4380202.py", line 39, in compute_communication_complexity
    if any(point & (1 <= k) != 0 for k in points[j]):
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

TypeError: 'int' object is not iterable

Judge reasoning

Test crashed due to type error; unable to evaluate conjecture validity | next: Fix data structure handling in compute_communication_complexity to handle integer inputs

Standard Young Tableau Count Exponential Gap in Symmetric Power Decompositions

- **Verdict:** INCONCLUSIVE
- **Bridge:** Algebraic Combinatorics (Standard Young Tableaux) $\times$ Monotone Circuit Size for Permanent
- **Recorded:** 2026-05-10 09:21 UTC
- **Entry ID:** 04216a9a3c0d

Statement

For every $n \geq 2$, the number of standard Young tableaux of shape $\lambda = (n-1, 1)$ for the permanent polynomial exceeds the number for the determinant polynomial by a factor of $\Omega(2^n)$.

Rationale

The shape $\lambda = (n-1, 1)$ corresponds to the dominant weight of the symmetric power representation, which captures the combinatorial structure of monomial expansions. The exponential gap in tableau counts reflects the inherent complexity of symmetric vs alternating decompositions in monotone circuits.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_11ee74bb.py", line 48, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1lee74bb.py", line 31, in run_trial
    ratio = math.log2(T_permanent / T_determinant)
                        ~~~~~^~~~~~
ZeroDivisionError: division by zero
```

Judge reasoning

Test crashed with ZeroDivisionError, preventing result evaluation. No counterexample found but support fraction cannot be assessed. | next: Fix division-by-zero handling in test code and re-run with diverse seeds